

Zihao Zhu

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📄 Google Scholar (1000+ Citations)

Education

The Chinese University of Hong Kong, Shenzhen <i>Ph.D. in Data Science, School of Data Science</i> Advisor: Baoyuan Wu	Shenzhen, China 2021–2026
University of Chinese Academy of Sciences <i>M.S. in Cybersecurity, Institute of Information Engineering</i> Advisor: Jing Yu	Beijing, China 2018–2021
China University of Mining and Technology <i>B.S. in Computer Science and Technology, School of Computer Science and Technology</i>	Xuzhou, China 2014–2018

Research Interests

Large Language Model Safety, Reasoning Models, Embodied AI Agents, Data Governance, AI Security & Privacy, Multimodal Learning

Research Experience

University at Buffalo, SUNY <i>Postdoctoral Researcher</i>	Buffalo, NY, USA 2026–Present
○ Continuing research on trustworthy AI, large language models, and AI safety.	
Tencent AI Lab <i>Research Intern</i>	Shenzhen, China Mar. 2024–July 2024
○ Led the design of EARBench, the first physical risk awareness evaluation benchmark for embodied AI. Built EARDataset using multi-agent collaboration to generate risk scenarios for task planning safety evaluation.	

Publications

Published/accepted 20 journal, conference, and workshop papers (8 as first author); 3 preprints (2 as first author)

(TPAMI: 3; IJCV: 2; Neurocomputing: 1; NeurIPS: 4; ICLR: 3; CVPR: 1; IJCAI: 2; AAAI: 1; ICASSP: 2; PR: 1)

Large Language Model Safety & Chain-of-Thought

- To Think or Not to Think: Exploring the Unthinking Vulnerability in Large Reasoning Models
Zihao Zhu, Hongbao Zhang, Ruotong Wang, Xu Ke, Siwei Lyu, Baoyuan Wu (NeurIPS 2025 Workshop)
- Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers
Zihao Zhu contributed as one of the project participants (NeurIPS 2025 Workshop)
- AdvChain: Adversarial Chain-of-Thought Tuning for Robust Safety Alignment of Large Reasoning Models

Zihao Zhu, Xinyu Wu, Gehan Hu, Siwei Lyu, Ke Xu, Baoyuan Wu (ICLR 2026)

AI Agents.....

4. Stealth Apart, Harm Together: Skill Cascading Attacks on Skill-Based Agent Systems
Zihao Zhu, Siwei Lyu, Adel Bibi, Baoyuan Wu (NeurIPS 2026)
5. EARBench: Towards Evaluating Physical Risk Awareness for Task Planning of Foundation Model-based Embodied AI Agents
Zihao Zhu, Bingzhe Wu, Zhengyou Zhang, Lei Han, Qingshan Liu, Baoyuan Wu (arXiv 2024)
6. BrandFusion: A Multi-Agent Framework for Seamless Brand Integration in Text-to-Video Generation
Zihao Zhu, Ruotong Wang, Siwei Lyu, Min Zhang, Baoyuan Wu (CVPR 2026 Findings)

Data Governance.....

7. VDC: Versatile Data Cleanser based on Visual-Linguistic Inconsistency by Multimodal Large Language Models
Zihao Zhu, Mingda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu (ICLR 2024)
8. HMGIE: Hierarchical and Multi-Grained Inconsistency Evaluation for Vision-Language Data Cleansing
Zihao Zhu, Hongbao Zhang, Guanzong Wu, Siwei Lyu, Baoyuan Wu (arXiv 2024)

AI Security.....

9. Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective
Baoyuan Wu, Zihao Zhu, Li Liu, Qingshan Liu, Zhaofeng He, Siwei Lyu (IJCV 2025)
10. Defenses in Adversarial Machine Learning: A Survey
Baoyuan Wu, Mingli Zhu, Meixi Zheng, Zihao Zhu, Shaokui Wei, Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu (IEEE TPAMI 2025)
11. BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks
Meixi Zheng, Xuanchen Yan, Zihao Zhu, Hongrui Chen, Baoyuan Wu (IEEE TPAMI 2025)
12. BackdoorBench: A Comprehensive Benchmark and Analysis of Backdoor Learning
Baoyuan Wu, Hongrui Chen, Mingda Zhang, Zihao Zhu, Shaokui Wei, Danni Yuan, Mingli Zhu, Ruotong Wang, Li Liu, Chao Shen (IJCV 2025)
13. Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers
Ruotong Wang, Hongrui Chen, Zihao Zhu, Li Liu, Baoyuan Wu (IEEE TPAMI 2024)
14. Reliable Poisoned Sample Detection against Backdoor Attacks Enhanced by Sharpness Aware Minimization
Mingda Zhang, Mingli Zhu, Zihao Zhu, Baoyuan Wu (ICLR 2026)
15. BackdoorBench: A Comprehensive Benchmark of Backdoor Learning
Baoyuan Wu, Hongrui Chen, Mingda Zhang, Zihao Zhu, Shaokui Wei, Danni Yuan, Hongyuan Zha (NeurIPS 2022)
16. Boosting Backdoor Attack with a Learnable Poisoning Sample Selection Strategy
Zihao Zhu, Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu (Neurocomputing 2026)

Multimodal Learning.....

17. From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering
Zihao Zhu (ICASSP 2022)

18. Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering
Zihao Zhu, Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu (IJCAI 2020, 177 citations)
19. Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering
Jing Yu, **Zihao Zhu**, Yujing Wang, Weifeng Zhang, Yue Hu, Jianlong Tan (Pattern Recognition 2020)

Other Publications

20. Learning to Optimize Permutation Flow Shop Scheduling via Graph-based Imitation Learning
Longkang Li, Siyuan Liang, **Zihao Zhu**, Xiaochun Cao, Chris Ding, Hongyuan Zha, Baoyuan Wu (AAAI 2024)
21. Enhanced Few-Shot Class-Incremental Learning via Ensemble Models
Mingli Zhu, **Zihao Zhu**, Sihong Chen, Chen Chen, Baoyuan Wu (arXiv 2024)
22. MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension
Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, **Zihao Zhu**, Yajing Sun (ICASSP 2021)
23. DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses
Xiaoze Jiang, Jing Yu, Zengchang Qin, **Zihao Zhu**, Qi Wu (IJCAI 2020)

Academic Service

Reviewer: IEEE TIP, ICLR, ICML, NeurIPS, CVPR, ICCV, AAAI, AISTATS

Awards & Honors

2024: Guotai Junan Scholarship

2021: AIRS Talent of PhD Research Program

2020: University of Chinese Academy of Sciences Merit Student

2018: Outstanding Undergraduate Thesis Award

2015, 2016: The First Prize Scholarship of CUMT